CSE 108: Object Oriented Programming I Sessional

Online 4: Introduction to Java

Total marks: 10 **Section: B1/B2** **Time: 35 minutes**

You will implement a simple **Gym Member** and **Late Fee Calculation System** using Java. The number of **members** and **subscription plans** are provided as **command line arguments**.

Requirements:

1. **Classes to Implement:** Implement the following classes:
   - Plan.java
   - GymMember.java
   - Main.java

2. **Plan Input:** Take details of subscription plans, including:
   a. Plan name (String)
   b. Late fee per day (double)

3. **Member Input:** Collect gym member information, including:
   - Member name (String)
   - Number of late-payment days for each plan (int)

   Ensure that:

   - Late days must be greater than or equal to 0
   - If invalid input is given, take the input again

4. **Late Fee Calculation:** Fine for each plan is calculated as:

   Late Fee = late_days × late_fee_per_day

5. **Total Late Fee:** Compute the **total late fee** for each member by summing fees from all plans.
   Note:
   - Late fee calculation must be done inside the **GymMember** class
   - Use appropriate getters and setters to access and modify class fields

6. **User Interaction:** After taking all inputs, the user can enter the following commands:
   - display X: Shows details for member X. Details include: *Member name, late fee for each plan, total late fee*. If X is "all", display details for all members.

- delete X: Deletes the information for member X. You may use a new array to store the updated member information.
- exit: Terminates the program.

7. **Additional Notes**

- Use arrays only (do not use ArrayList or other collections)
- The Main class should only handle user input and command processing
- All calculations must be done using basic Java constructs (loops, arrays, conditionals)

Submission

Create a directory with your ID and include only the .java files. For example:

```
2405072/
├── Plan.java
├── GymMember.java
└── Main.java
```

Submit the folder as:
2405072.zip